Ex.No.-1.2-Design and Draft the given 2D Sketches in modelling software.

AIM:

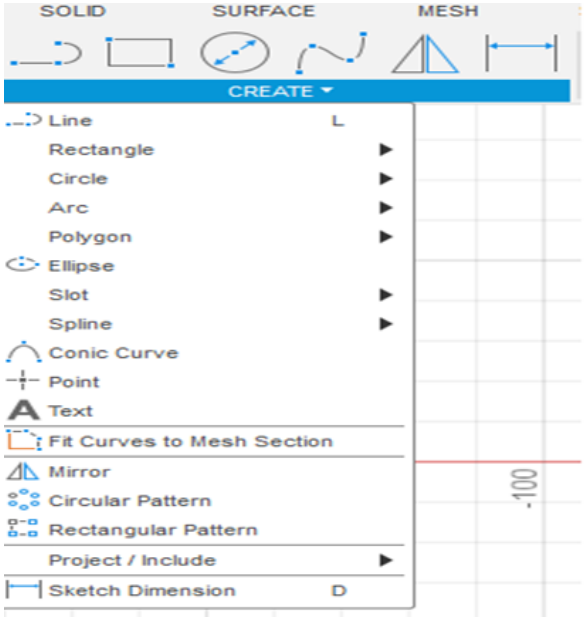
To sketch the given drawing with dimensions using fusion 360 tool.

REQUIREMENTS:

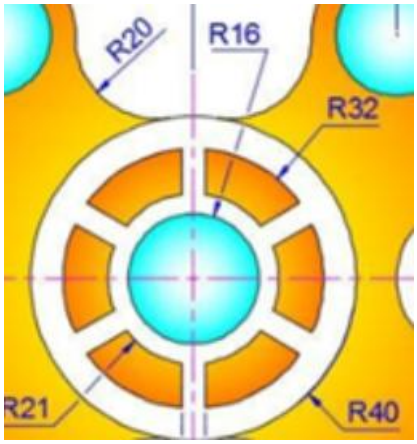
1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

1. Open Fusion 360, click Create Sketch, and pick the Right plane.



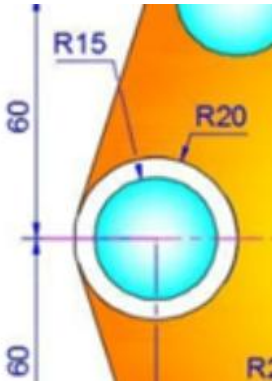
2. Draw the center circles with radius 16 and radius 40 at the middle point



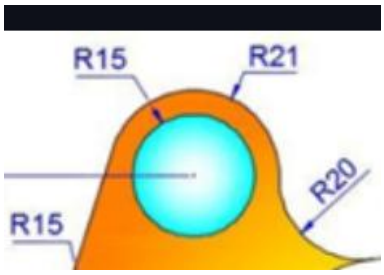
3. Draw the inner ring cuts and the vertical 6 mm gap down the center.



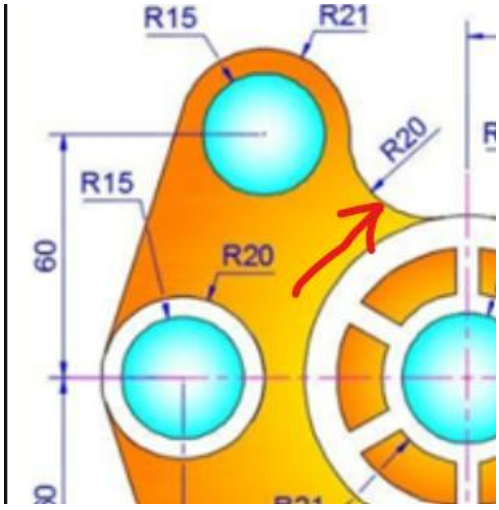
4. Move 50 mm left and right to draw the middle holes (radius 15) and outer curves (radius 20).



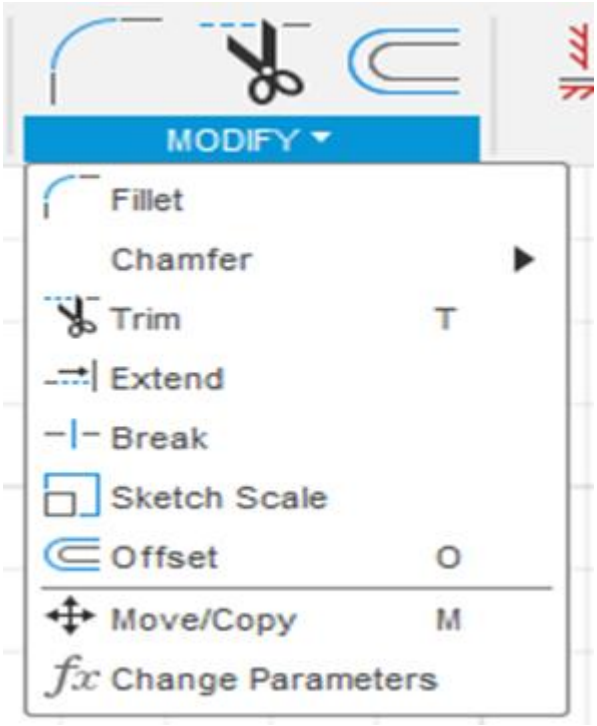
5. Move 50 mm left/right and 60 mm up to draw the top holes (radius 15) and outer curves (radius 21).



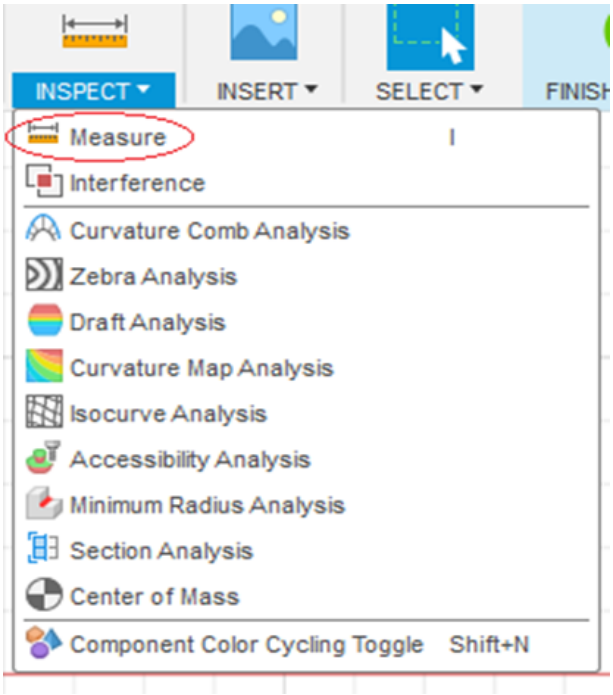
6. Move 50 mm left/right and 60 mm down to draw the bottom holes (radius 15) and outer curves (radius 21).
7. Connect the outer shapes using radius 20 arcs and straight side lines.(bottom and top)



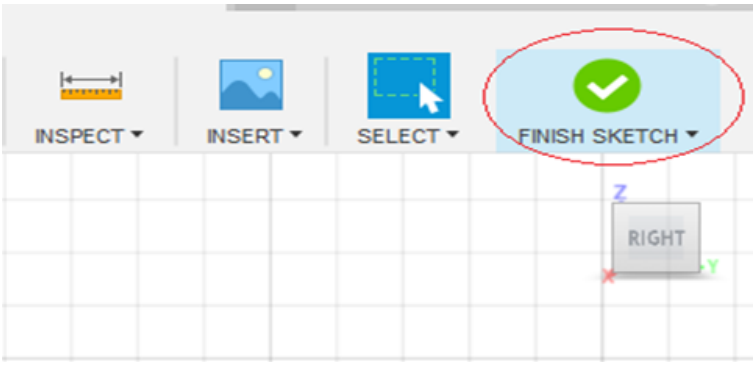
8. Click Modify, choose Trim (or press T), and cut away all extra lines inside the shape.



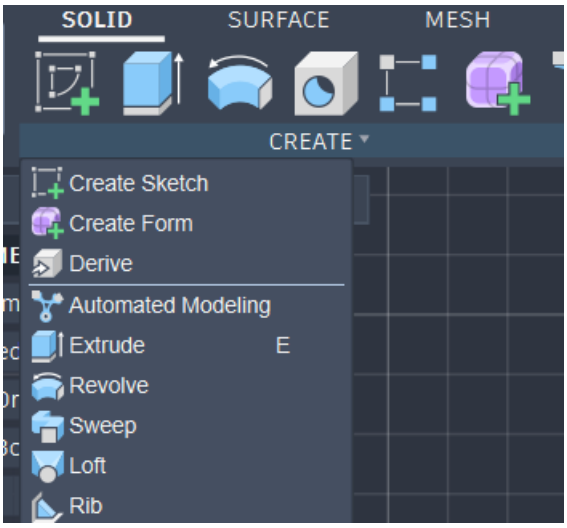
9. Use inspect feature to measure dimensions of sketch.



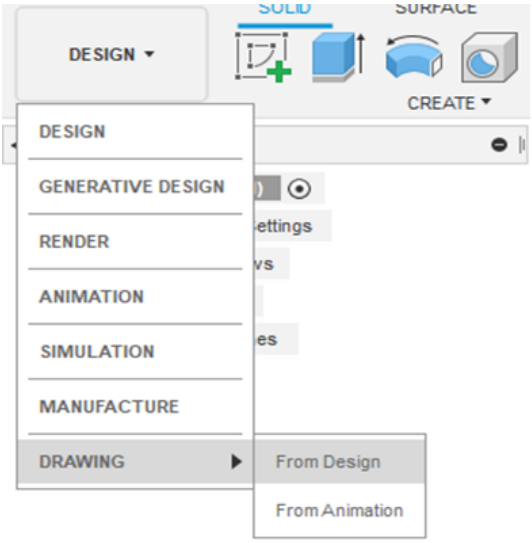
10. After finishing the sketch use finish sketch option to finalize the sketch.

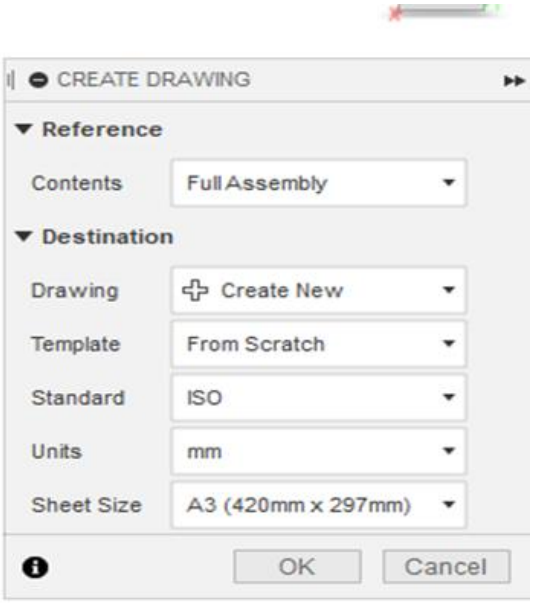


11. After finishing the sketch, under solid under create, click on extrude and set the value to 20mm.

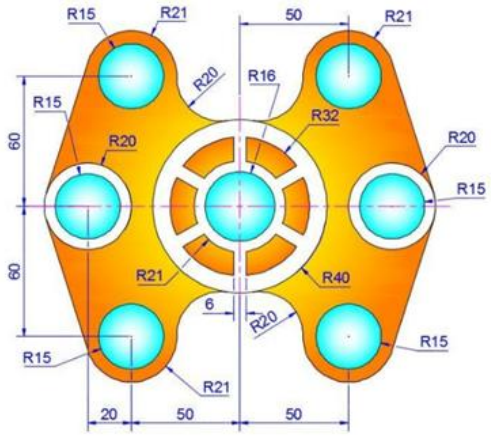


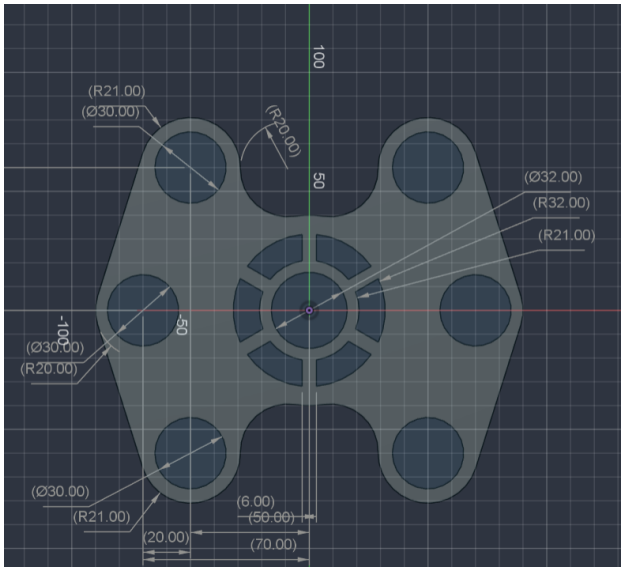
12. After finishing the sketch switch to drawing mode to plot the sketch in ISO standard.



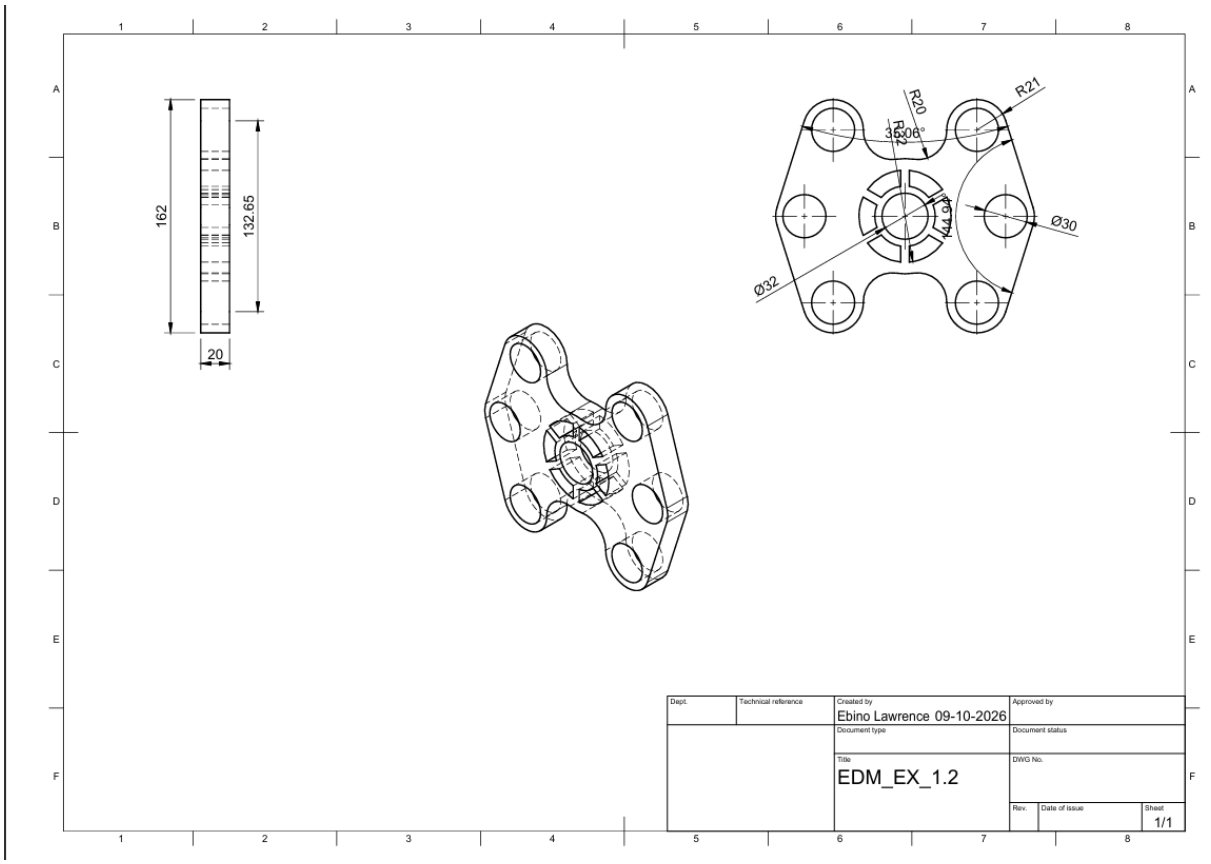

CREATE DRAWING
Reference
Contents
Full Assembly
Destination
Drawing
Create New
Template
From Scratch
Standard
ISO
Units
mm
Sheet Size
A3 (420mm x 297mm)
OK
Cancel

SKETCH:


R15
R21
50
R21
R20
R16
R15
R32
R20
R20
R15
60
60
R21
R40
6
R20
R15
R15
R21
20
50
50


100
(R21.00)
(Ø30.00)
(R20.00)
50
(Ø32.00)
(R32.00)
(R21.00)
-100
-50
(Ø30.00)
(R20.00)
(Ø30.00)
(R21.00)
(6.00)
(50.00)
(20.00)
(70.00)

OUTPUT:



RESULT:

Thus, the given sketch is drawn and drafted using fusion 360 tool.